MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

RECEIVER, TRANSLATE, DIGITAL, KEY CODE MDQ

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic translate receiver module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 2.9 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance and requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be MDQ.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/51-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

* 3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltages of +15.0 ±1.5 V dc, -15.0 ±1.5 V dc, and -5.0 ±0.5 V dc.

4. QUALITY ASSURANCE PROVISIONS

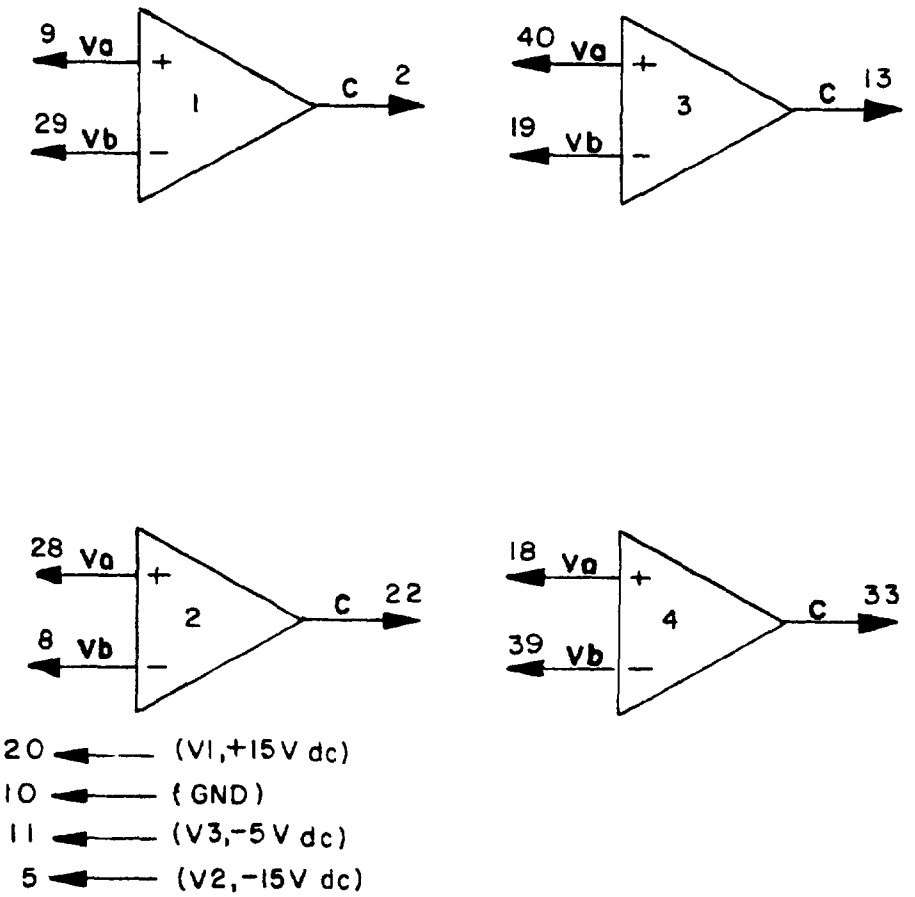
4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

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NOTE: Unused pins: 1, 3, 4, 6, 7, 12, 14, 15, 16, 17, 21, 23, 24, 25, 26, 27, 30, 31, 32, 34, 35, 36, 37, and 38.

* FIGURE 1. MDQ module function diagram, translate receiver.

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* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/ 3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1	Isolation	III	S	10	∞	1	∞	10	∞	1	∞	MΩ
A2-A25	Isolation	III	Q	10	∞	1	∞	10	∞	1	∞	MΩ
B1-B4	Input capacitance	IV	Q	0.008	0.012	0.008	0.012	0.008	0.012	0.008	0.012	µF
C1-C4	Output voltage, low level	V	X	-1.0	0.4	-1.0	0.4	-1.0	0.4	-1.0	0.4	V
D1-D4	Output voltage, high level	V	X	2.4	5.0	2.4	5.0	2.4	5.0	2.4	5.0	V
E1-E12	Common mode rejection, low level	VI	S	-1.0	0.4	-1.0	0.4	-1.0	0.4	-1.0	0.4	V
F1-F8	Common mode rejection, high level	VII	S	2.4	5.0	2.4	5.0	2.4	5.0	2.4	5.0	V
G1-G4	Input current, low level	VIII	X	0.2	0.5	0.2	0.5	0.2	0.5	0.2	0.5	mA
H1-H4	Input current, high level	VIII	X	0.8	1.5	0.8	1.5	0.8	1.5	0.8	1.5	mA
J1-J4	Input current to (-) inputs	IX	Q	3.9	4.5	3.9	4.5	3.9	4.5	3.9	4.5	mA
J5-J8	Input current to (-) inputs	IX	Q	3.5	4.0	3.5	4.0	3.5	4.0	3.5	4.0	mA
K1-K8	Hysteresis	X	Q	200	400	200	400	200	400	200	400	mV
L1	Power supply current, low level, +15 V dc	XI	S	20	64	20	64	20	64	20	64	mA
L2	Power supply current, low level, -15 V dc	XI	S	6	18	6	18	6	18	6	18	mA
L3	Power supply current, low level, -5 V dc	XI	S	12	40	12	40	12	40	12	40	mA
L4	Power supply current, high level, +15 V dc	XI	S	20	64	20	64	20	64	20	64	mA
L5	Power supply current, high level, -15 V dc	XI	S	8	20	8	20	8	20	8	20	mA
L6	Power supply current, high level, -5 V dc	XI	S	10	50	10	50	10	50	10	50	mA

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
M1-M4	Transition time (t_{TLH})	XII	S	5	75	5	75	5	75	5	75	ns
N1-N4	Transition time (t_{THL})	XII	S	10	125	10	125	10	125	10	125	ns
P1-P4	Propagation delay time (t_{PLH})	XII	S	15	50	15	50	15	55	15	55	ns
R1-R4	Propagation delay time (t_{PHL})	XII	S	15	50	15	50	15	55	15	55	ns
S1-S3	Power supply transient amplitude	XIII	Q	0	400	0	400	0	400	0	400	mV
T1-T4	Output crosstalk, low level	XIV	Q	0	400	0	400	0	400	0	400	mV
U1-U4	Output crosstalk, high level	XIV	Q	0	400	0	400	0	400	0	400	mV
V1	Power supply transient rise time	XV	S	1.4	2.5	1.4	2.5	1.4	2.5	1.4	2.5	ms
V2	Power supply transient rise time	XV	S	5	12	5	12	5	12	5	12	µs
V3	Power supply transient rise time	XV	S	1.2	2.0	1.2	2.0	1.2	2.0	1.2	2.0	ms

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through V shall be performed in any order with the following exceptions. If test conditions A is part of a sequence of tests, the order of testing shall be as follows: A where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through V in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C through R and V shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

* 4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition V, shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests C through V shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests A through V shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Military specification MIL-M-28787/51 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No 10001-2833425. Part number M28787/51-1 supersedes part numbers 2833425 and M28787/51

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* 6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Supply voltage	+15.0 ±1.5 V dc -15.0 ±1.5 V dc -5.0 ±0.5 V dc
Maximum power dissipation	
All outputs at low level	1.35 W
All outputs at high level	1.45 W
Output voltage	
Low level	-1.0 V minimum, 0.4 V maximum
High level	2.4 V minimum, 5.0 V maximum
Output current	
Low level (V_{OL} = 0.4 V)	1 73 mA maximum
High level (V_{OH} = 2.4 V)	100 µA maximum
Propagation delay time	
t_{PLH}	55 ns maximum
t_{PHL}	55 ns maximum
Power supply current (outputs not loaded)	
+15 V dc	64 mA maximum
-15 V dc	20 mA maximum
-5 V dc	50 mA maximum
Input threshold	
Low Level	When the voltage at the plus (+) input (Va) is -1.9 V dc, or more negative than -1.9 V dc, with respect to the voltage at the minus (-) input (Vb), the input is recognized as a low level; Va-Vb $\geq$ - 1.9 V dc.
High level	When the voltage at the plus (+) input (Va) is -1.1 V dc, or less negative than -1.1 V dc, with respect to the voltage at the minus (-) input (Vb), the input is recognized as a high level; Va-Vb $\leq$ -1.1 V dc.
Input current	
Low level (Va = -3 V dc, Vb = 0 V dc)	0.5 mA maximum
High level (Va = Vb = 0 V dc	-1.5 mA maximum
Common mode rejection	±7.5 V

* 6.5 Module function. This module contains four negative interface line receiver circuits designed to translate logic signals from negative voltage levels to positive voltage levels. The MDQ module is intended to be used with translate driver module MDP (MIL-M-28787/48), or equivalent. The output will be a low level when both inputs are open. The input circuit presents a terminal impedance to the line of approximately 150Ω in series with 0.01 µF. When used with twisted shielded pair, the high side of the line should be connected to the plus (+) input, the low side of the line should be connected to the minus (-) input, and the shield should be connected to the chassis.

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6.5.1 Input to output relationship. Each translate receiver has the input to output relationship of table II for positive logic.

* 6.5.2 Parallel outputs. Paralleling of outputs is not permissible.

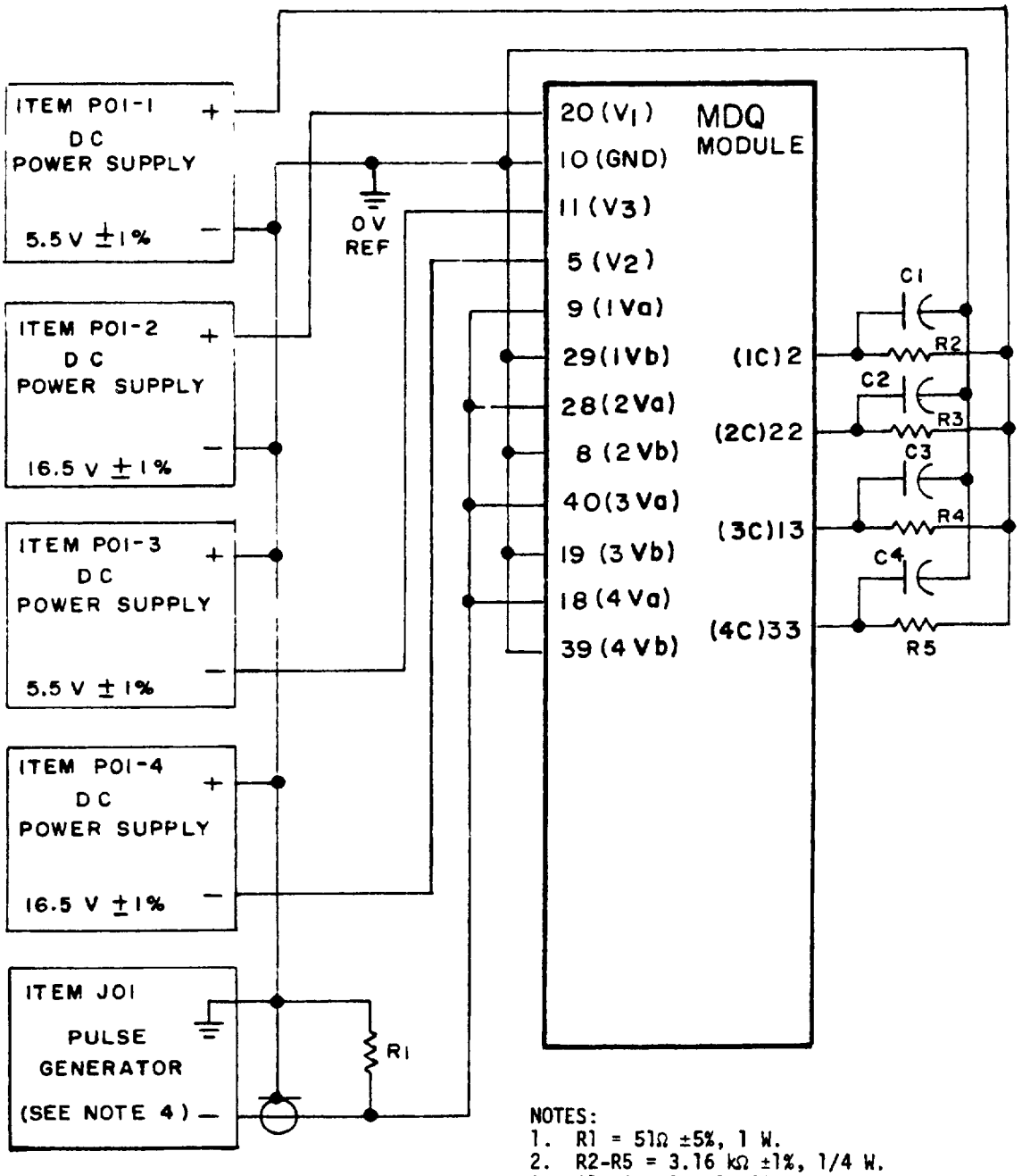
6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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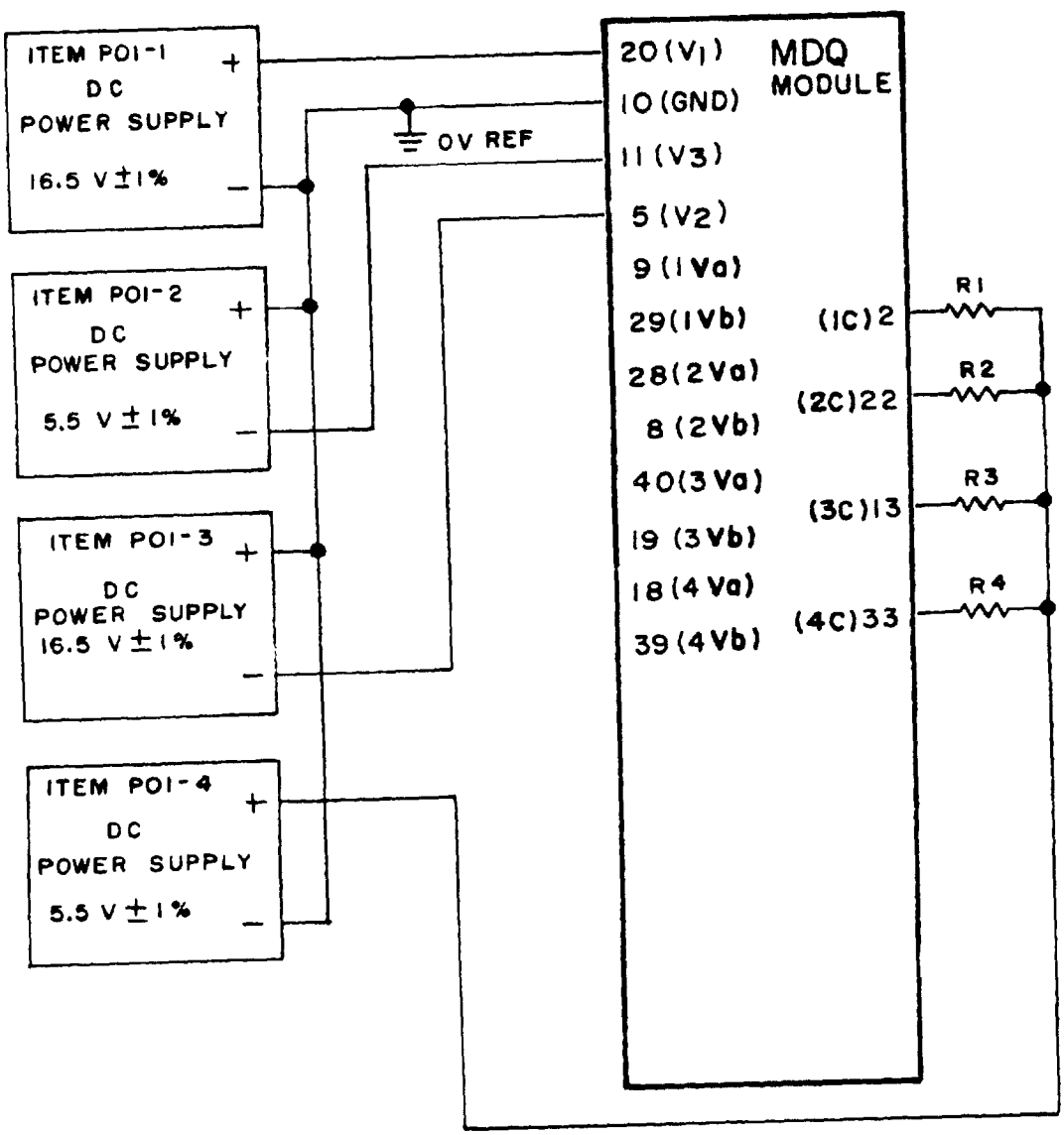
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NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. R2-R5 = 3.16 kΩ ±1%, 1/4 W.
3. C1-C4 = 50 pF ±20%.
4. Item J01 input pulse parameters:
   PA = -3.0 V ±10%.
   Up level = 0.0 ±0.2 V.
   Down level = -3.0 V ±10%.
   PW = 1.0 µs ±10%
   PRR = 0.5 MHz ±10%.
   $t_{TLH} = t_{THL} \leq 100$ ns

* FIGURE 2. Operating temperature and life test circuit.

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NOTE: R1-R4 = 3.16 kΩ ±1%, 1/4 W

* FIGURE 3. Component temperature test circuit.

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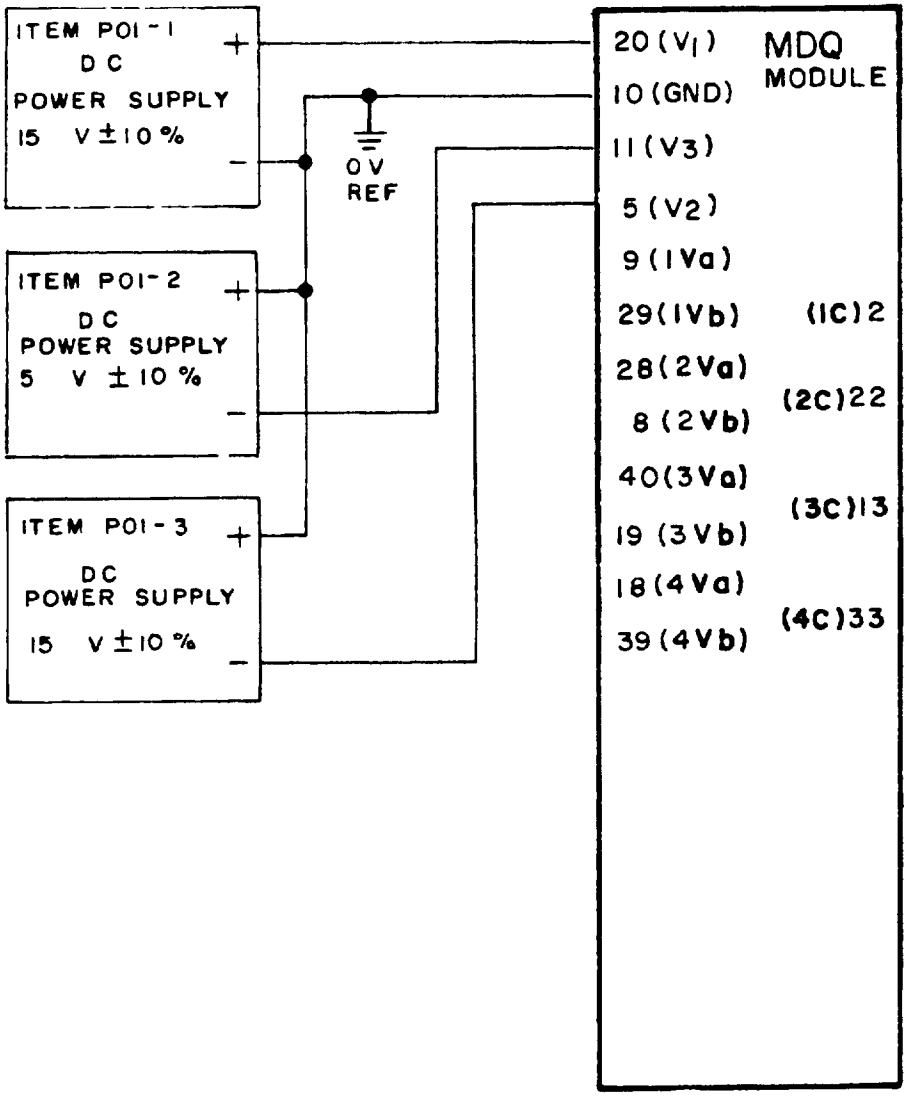


FIGURE 4. Vibration test circuit.

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TABLE I. Electrical test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation	C,1,3,4,6,7,12,14,15,16,17,21,23,24,25,26,27,30,31,32,34,35,36,37,38	Frame	10	∞	MΩ	5
A2		C	1,3,4,6,7,12,14,15,16,17,21,23,24,25,26,27,30,31,32,34,35,36,37,38				
A3		1	3,4,6,7,12,14,15,16,17,21,23,24,25,26,27,30,31,32,34,35,36,37,38				
A4		3	4,6,7,12,14,15,16,17,21,23,24,25,26,27,30,31,32,34,35,36,37,38				
A5		4	6,7,12,14,15,16,17,21,23,24,25,26,27,30,31,32,34,35,36,37,38				
A6		6	7,12,14,15,16,17,21,23,24,25,26,27,30,31,32,34,35,36,37,38				
A7	▼	7	12,14,15,16,17,21,23,24,25,26,27,30,31,32,34,35,36,37,38	▼	▼	▼	▼

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TABLE III. Isolation test - Continued.

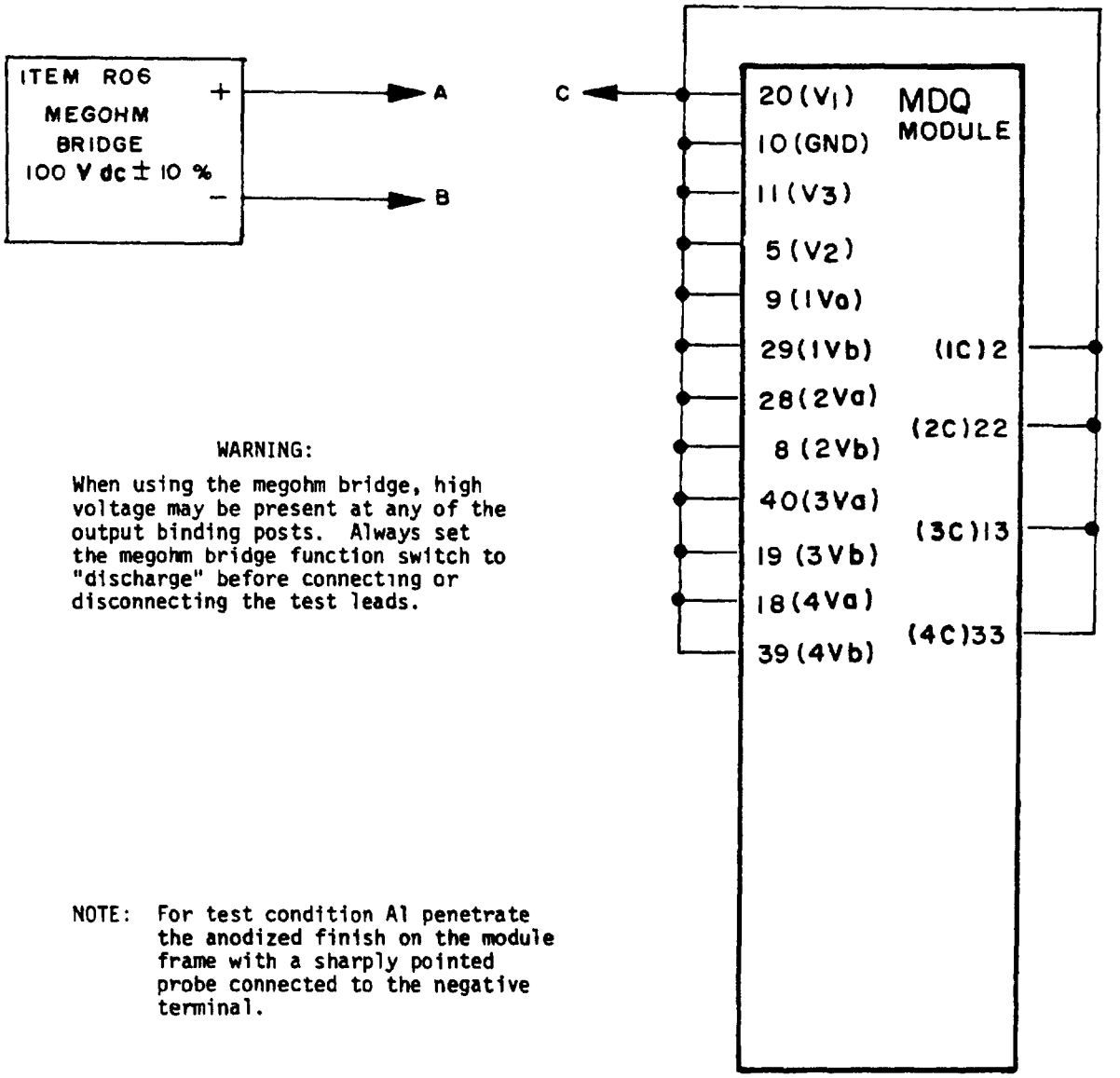
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A8	Isolation	12	14,15,16,17,21,23,24,25,26,27,30,31,32,34,35,36,37,38	10	∞	MΩ	5
A9		14	15,16,17,21,23,24,25,26,27,30,31,32,34,35,36,37,38				
A10		15	16,17,21,23,24,25,26,27,30,31,32,34,35,36,37,38				
A11		16	17,21,23,24,25,26,27,30,31,32,34,35,36,37,38				
A12		17	21,23,24,25,26,27,30,31,32,34,35,36,37,38				
A13		21	23,24,25,26,27,30,31,32,34,35,36,37,38				
A14		23	24,25,26,27,30,31,32,34,35,36,37,38				
A15		24	25,26,27,30,31,32,34,35,36,37,38				
A16	▼	25	26,27,30,31,32,34,35,36,37,38	▼	▼	▼	▼

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TABLE III. [illegible] continued.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A17	Isolation	26	27,30,31,32,34,35,36,37,38	10	∞	MΩ	5
A18		27	30,31,32,34,35,36,37,38				
A19		30	31,32,34,35,36,37,38				
A20		31	32,34,35,36,37,38				
A21		32	34,35,36,37,38				
A22		34	35,36,37,38				
A23		35	36,37,38				
A24		36	37,38				
A25	▼	37	38	▼	▼	▼	▼

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WARNING:

When using the megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: For test condition A1 penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 5. Isolation test circuit for table III.

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TABLE IV. Input capacitance.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	Input capacitance	9	29	0.008	0.012	µF	6
B2	↓	28	8	↓	↓	↓	↓
B3		40	19				
B4		18	39				

* TABLE V. Output voltage (low-level and high-level) test.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
C1	Output voltage, low level	B,29	A	9	Open	Open	2	-1.0	0.4	V	7
C2	↓	B,8	↓	28	↓	↓	22	↓	↓	↓	↓
C3		B,19		40			13				
C4		B,39		18			33				
D1	Output voltage, high level	B,29		9			2	2.4	5.0		
D2	↓	B,8		28			22	↓	↓		
D3		B,19		40			13				
D4		B,39		18			33				

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TABLE VI. Common mode rejection (low-level) test.

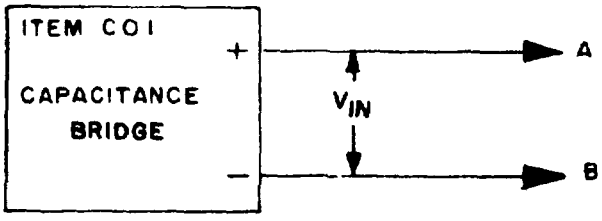
Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
E1	Common mode rejection, low level	C,E	9	A	29	A	2	-1.0	0.4	V	7
E2			28		8		22				
E3			40		19		13				
E4			18		39		33				
E5		B,D	A	9	A	29	2				
E6				28		8	22				
E7				40		19	13				
E8				18		39	33				
E9		Open	Open	Open	Open	Open	2				
E10							22				
E11							13				
E12							33				

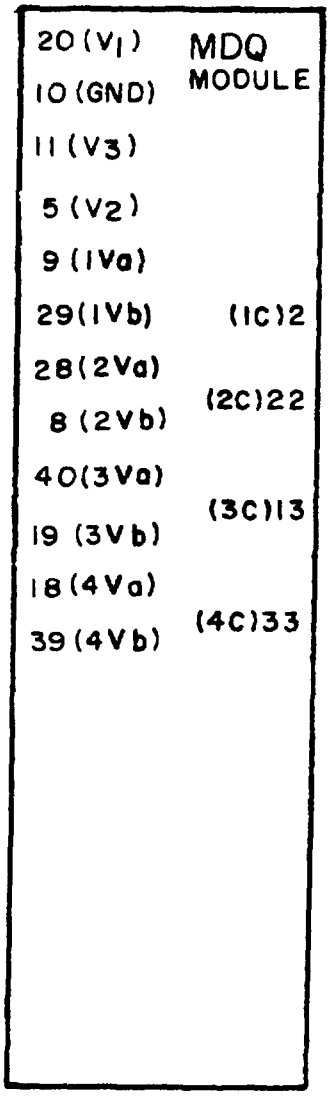
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TABLE VII. Common mode rejection (high-level) test.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
F1	Common mode rejection, high level	C,E	9	A	29	A	2	2.4	5.0	V	7
F2			28		8		22				
F3			40		19		13				
F4			18		39		33				
F5		B,D	A	9	A	29	2				
F6				28		8	22				
F7				40		19	13				
F8				18		39	33				

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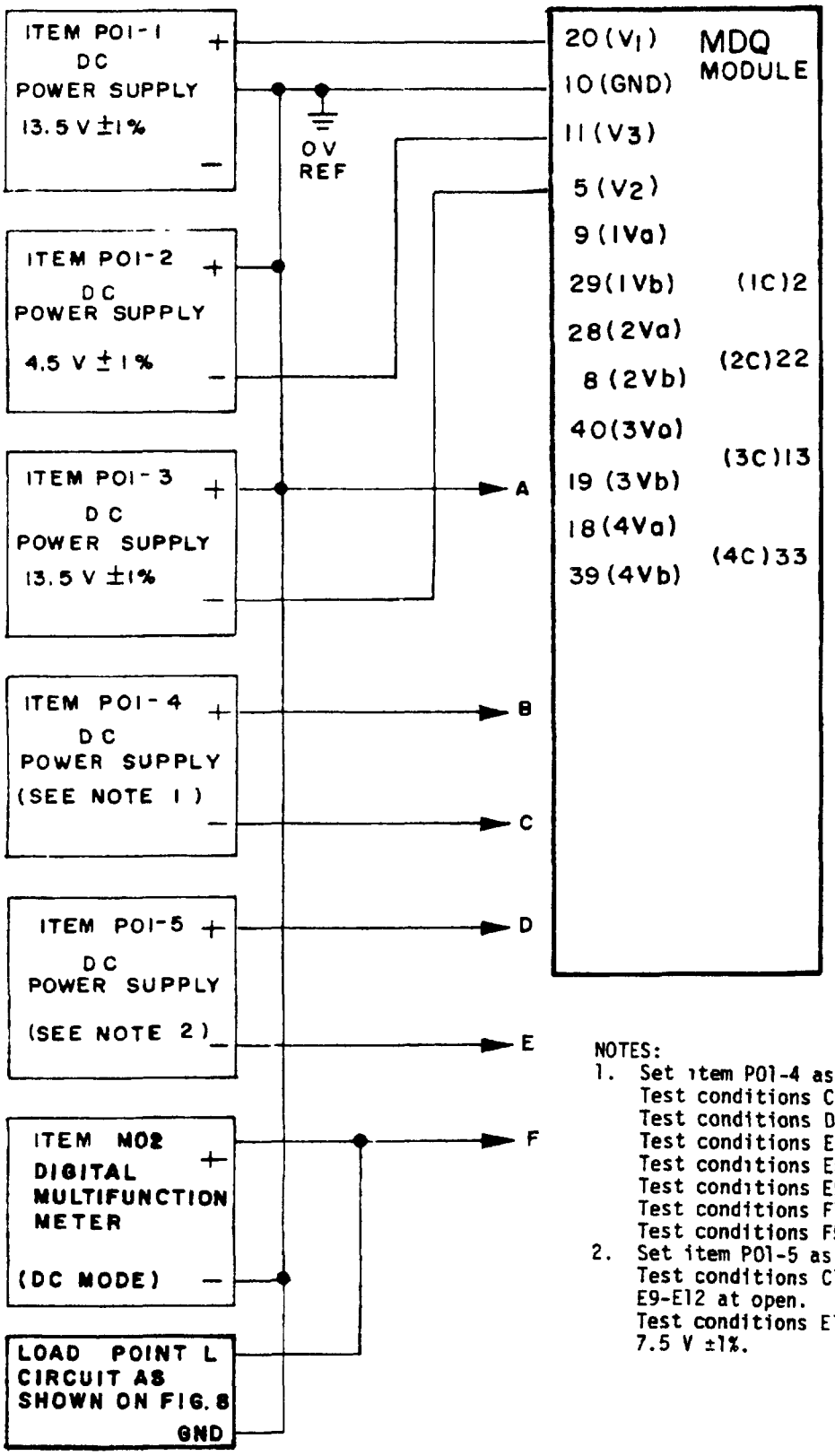




NOTES:
1. Set V_{IN} to 0.2 V at "unknown" terminals at null.
2. Measure the values of Q and G at 1.0 kHz.
3. Use formula $C_{IN} = \frac{QG}{2\pi f}$ to calculate value of input capacitance.

FIGURE 6. Input capacitance test circuit for table IV.

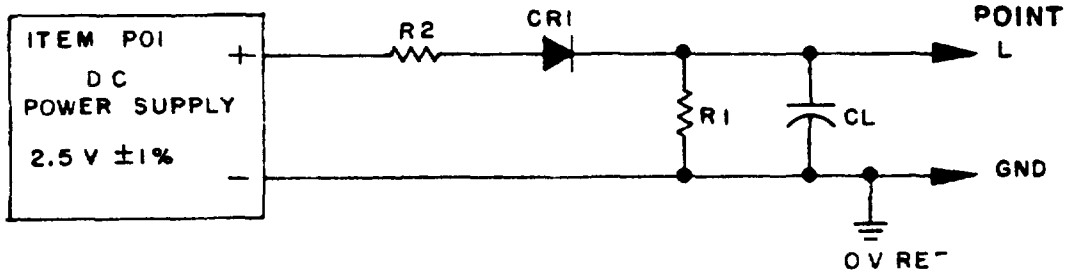
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NOTES:
1. Set item P01-4 as follows:
   Test conditions C1-C4 at 1.9 V ±1%.
   Test conditions D1-D4 at 1.1 V ±1%.
   Test conditions E1-E4 at 5.6 V ±1%.
   Test conditions E5-E8 at 9.4 V ±1%.
   Test conditions E9-E12 at open.
   Test conditions F1-F4 at 6.4 V ±1%.
   Test conditions F5-F8 at 8.6 V ±1%.
2. Set item P01-5 as follows:
   Test conditions C1-C4, D1-D4, and E9-E12 at open.
   Test conditions E1-E8 and F1-F8 at 7.5 V ±1%.

* FIGURE 7. Output voltage (low-level and high-level) and common mode rejection (low-level and high-level) test circuit for tables V, VI, and VII

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NOTES:
1. R1 = Select value for a current flow of 0.1 mA ±1% into point L when L is held at 2.4 Vdc.
2. R2 = Select value for a current flow of 1.73 mA ±1% from point L to ground.
3. C_L = 50 ±3 pF (including probe and parasitic capacitance).
4. CR1 = 1N4150, or equivalent.
5. C_L is not required for dc electrical tests C1-C4, D1-D4, E1-E12, and F1-F8.
6. The maximum lead length from point L to the appropriate module pin shall be 3 inches (76.2 mm).
7. During output switching, the 2.5 V power supply shall have a maximum overshoot of 500 mV.

FIGURE 8. Logic load circuit.

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* TABLE VIII. Input current (low-level and high-level) test.

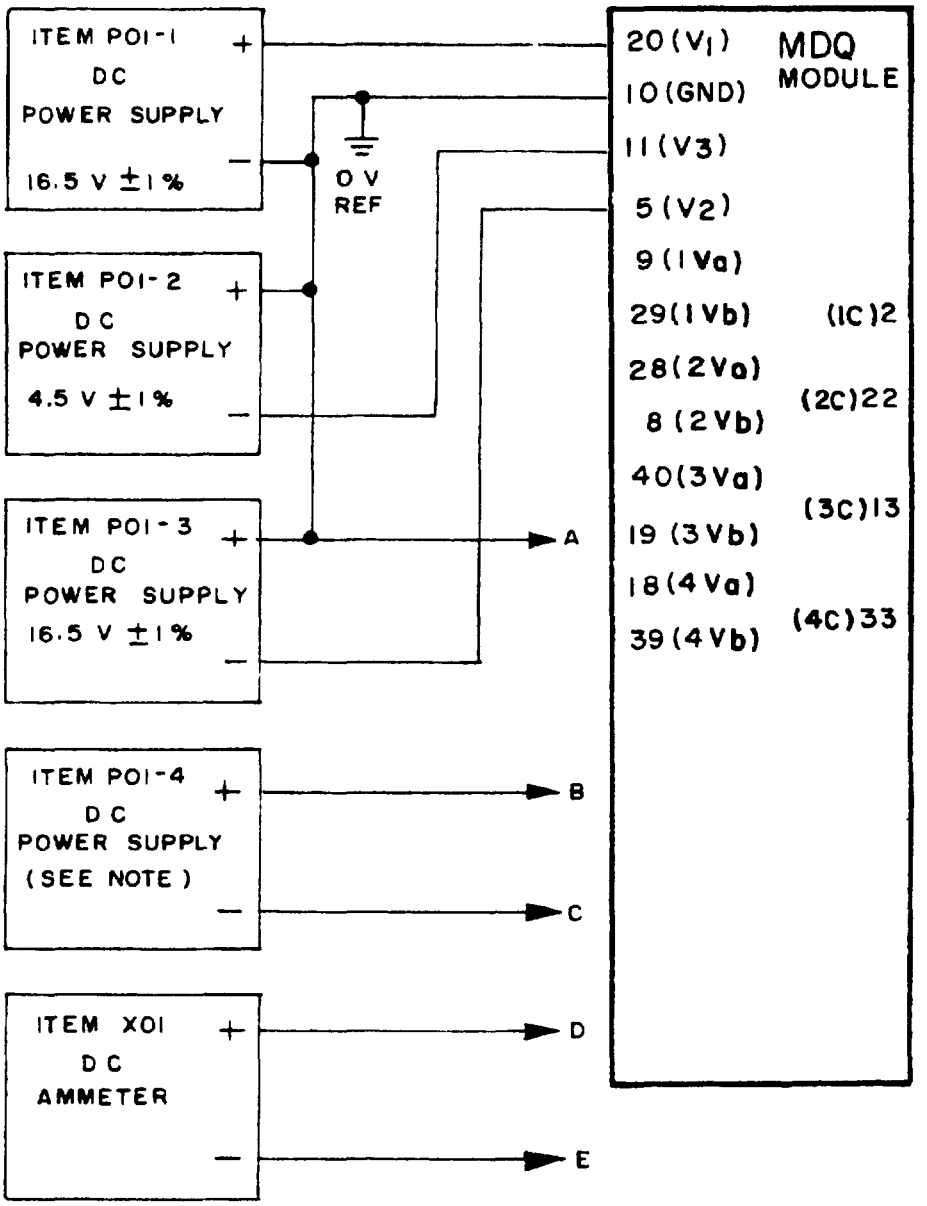
Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
G1	Input current, low level	29,B	A	E	9	C	0.2	0.5	mA	9
G2	↓	8,B	↓	↓	28	↓	↓	↓	↓	↓
G3	↓	19,B	↓	↓	40	↓	↓	↓	↓	↓
G4	↓	39,B	↓	↓	18	↓	↓	↓	↓	↓
H1	Input current, high level	29,D	Open	Open	A	9	0.8	1.5	↓	↓
H2	↓	8,D	↓	↓	↓	28	↓	↓	↓	↓
H3	↓	19,D	↓	↓	↓	40	↓	↓	↓	↓
H4	↓	39,D	↓	↓	↓	18	↓	↓	↓	↓

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TABLE IX. Input current to (-) inputs test.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
J1	Input current to (-) inputs	C	9,D	A	B	29	3.9	4.5	mA	9
J2		→	28,D	→	→	8	→	→	→	→
J3		→	40,D	→	→	19	→	→	→	→
J4		→	18,D	→	→	39	→	→	→	→
J5		B	A	9,E	29	C	3.5	4.0	→	→
J6		→	→	28,E	8	→	→	→	→	→
J7		→	→	40,E	19	→	→	→	→	→
J8		→	→	18,E	39	→	→	→	→	→

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTE. Set item P01-4 as follows:
Test conditions G1-G4 at 3.0 V ±1%.
Test conditions H1-H4 at open.
Test conditions J1-J8 at 7 5 V ±1%.

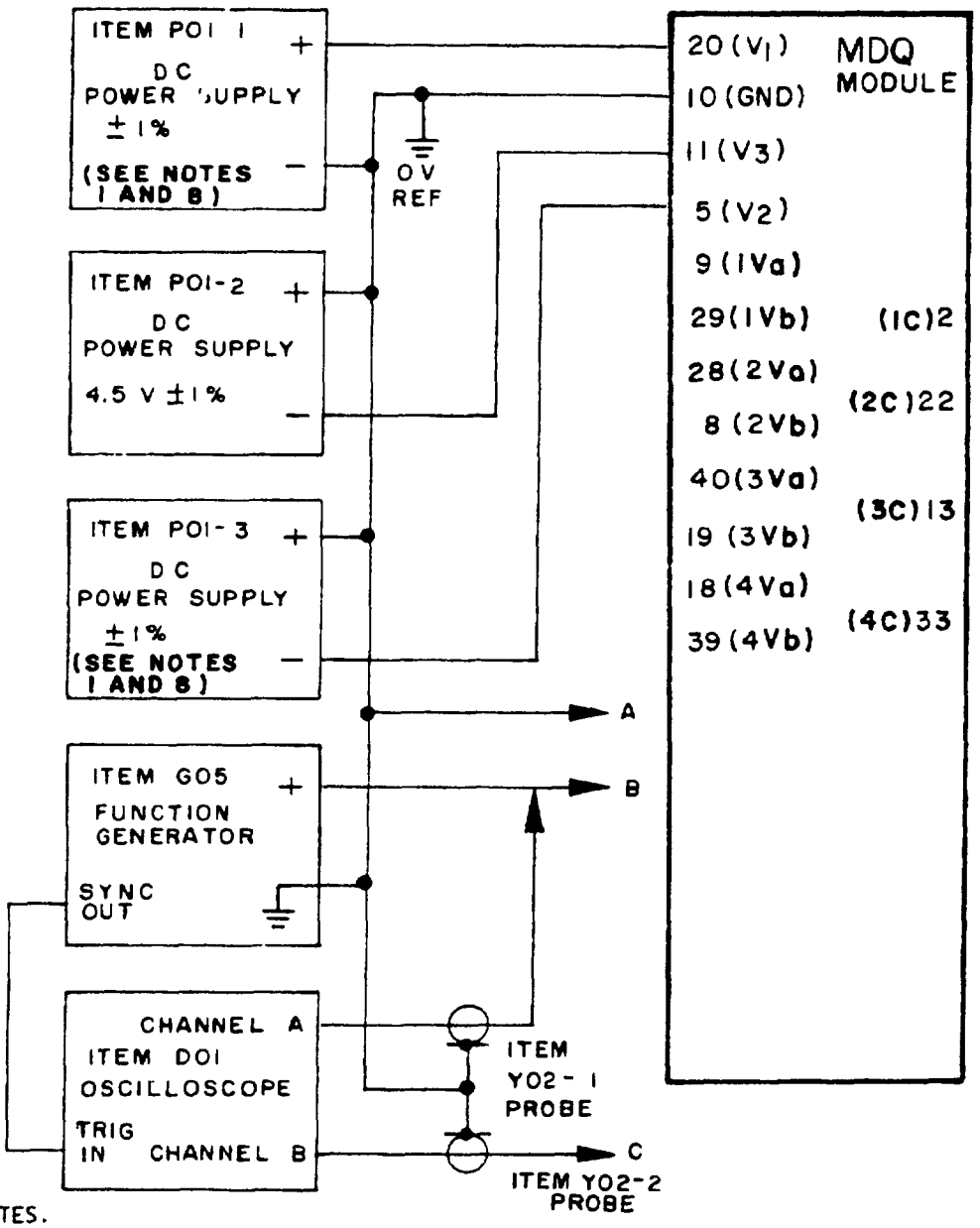
FIGURE 9. Input current (low-level and high-level) and input current to (-) inputs test circuit for tables VIII and IX.

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* TABLE X. Hysteresis test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
K1	Hysteresis	29	9	2	200	400	mV	10,11
K2	↓	8	28	22	↓	↓	↓	↓
K3		19	40	13				
K4		39	18	33				
K5		29	9	2				
K6		8	28	22				
K7		19	40	13				
K8	↓	39	18	33	↓	↓	↓	↓

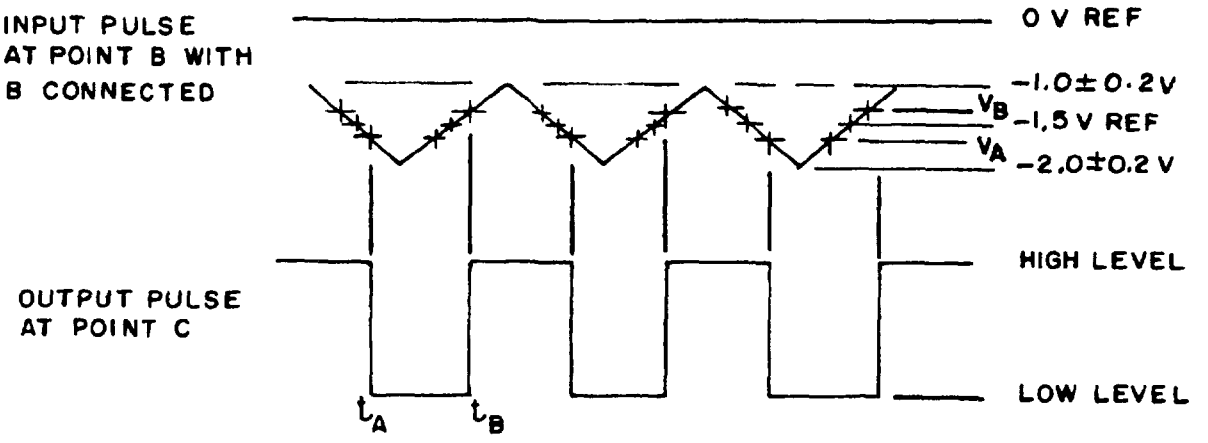
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NOTES.
1. Adjust items PO1-1 and PO1-3 to 13.5 V ±1%.
2. V_A and V_B are the voltage levels on the input pulse in time coincidence with output switching from high to low and low to high respectively The points t_A and t_B are the left and right most points respectively of the output low level.
3 Aline the low level of the output pulse with a horizontal marking on the scope face.
4. Aline V_A on the input pulse with the point t_A on the output pulse and note the position of the V_A level.
5. Aline V_B on the input pulse with the point t_B on the output pulse and note the position of the V_B level.
6. The result of $|V_A| - |V_B|$ (in cm) times the volt per centimeter setting of the input pulse shall comply with test condition K1.
7. Repeat notes 3 through 6 above for test conditions K2-K4
8. Adjust items PO1-1 and PO1-3 to 16 5 V ±1%.
9. Repeat notes 3 through 6 for test conditions K5-K8

* FIGURE 10. Hysteresis test circuit for table X.

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NOTE: Input pulse parameter:
PRR = 100 Hz ±10%.

FIGURE 11. Hysteresis test circuit waveforms for table X.

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* TABLE X. Power supply current and low- and high-level bias

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
L1	Power supply current, low level, +15 V dc	Open	9,28,40,18	20	64	mA	12
L2	Power supply current, low level, -15 V dc	Open	9,28,40,18	6	18		
L3	Power supply current, low level, -5 V dc	Open	9,28,40,18	12	40		
L4	Power supply current, high level, +15 V dc	9,28,40,18	Open	20	64		
L5	Power supply current, high level, -15 V dc	9,28,40,18	Open	8	20		
L6	Power supply current, high level, -5 V dc	9,28,40,18	Open	10	50		

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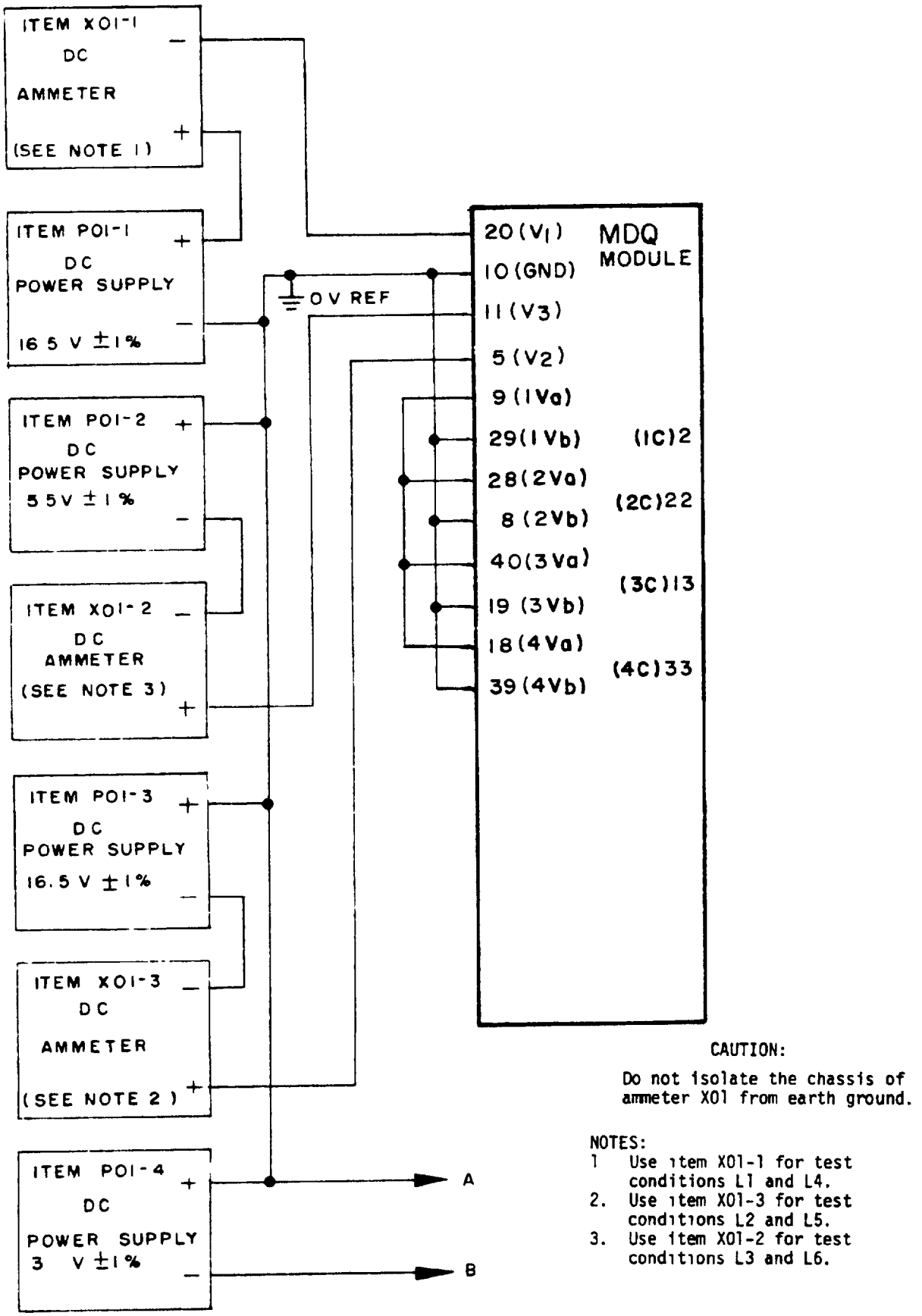


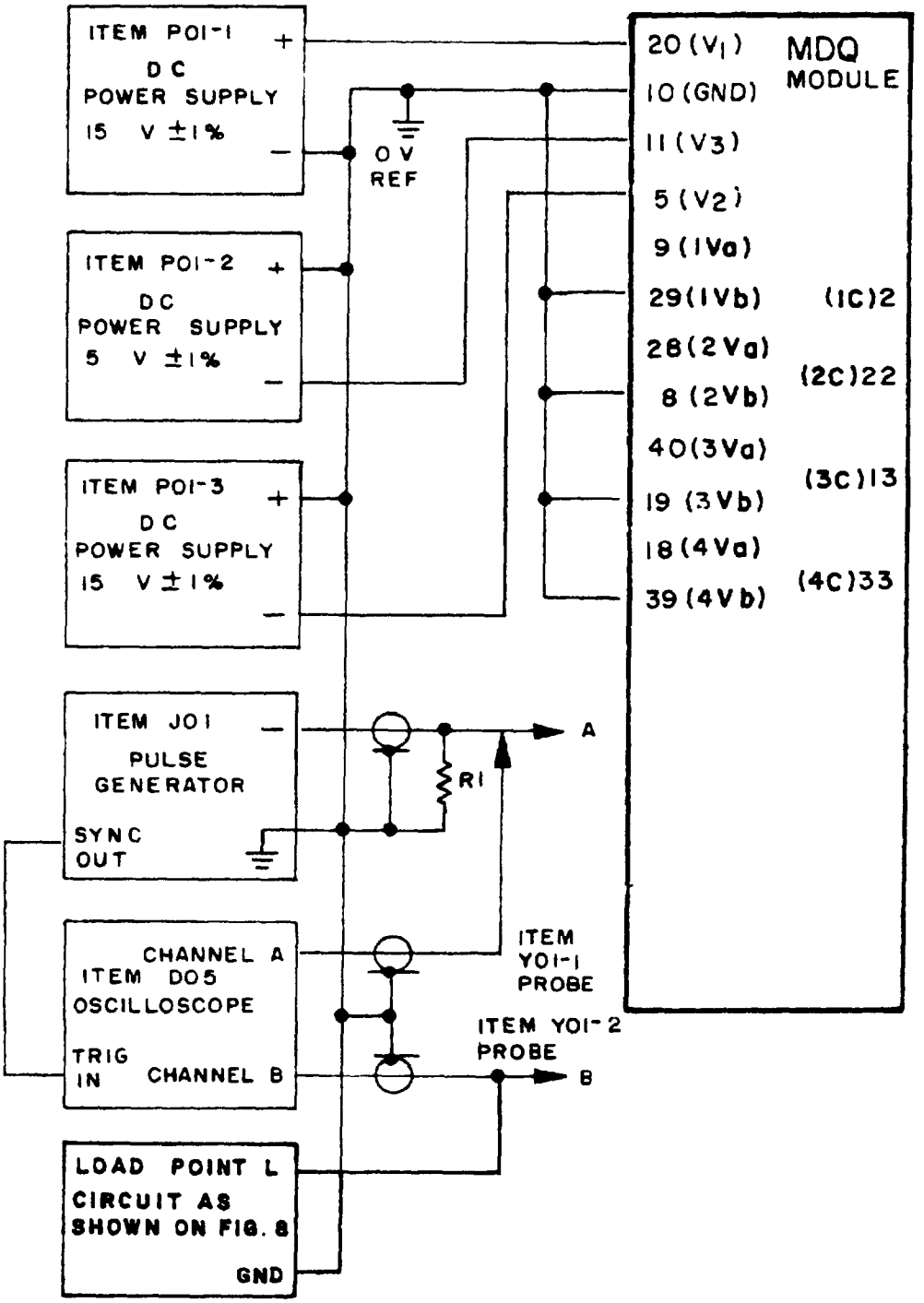
FIGURE 12. Power supply current (low-level and high-level) test circuit for table XI.

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TABLE XII. Transition time (t_{TLH} and [illegible] on delay time (t_{PLH} and [illegible]) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
M1	Transition time (t_{TLH})	9	2	5	75	ns	13,14
M2	↓	28	22	↓	↓	↓	↓
M3	↓	40	13	↓	↓	↓	↓
M4	↓	18	33	↓	↓	↓	↓
N1	Transition time (t_{THL})	9	2	10	125	↓	↓
N2	↓	28	22	↓	↓	↓	↓
N3	↓	40	13	↓	↓	↓	↓
N4	↓	18	33	↓	↓	↓	↓
P1	Propagation delay time (t_{PLH})	9	2	15	50	↓	↓
P2	↓	28	22	↓	↓	↓	↓
P3	↓	40	13	↓	↓	↓	↓
P4	↓	18	33	↓	↓	↓	↓
R1	Propagation delay time (t_{PHL})	9	2	↓	↓	↓	↓
R2	↓	28	22	↓	↓	↓	↓
R3	↓	40	13	↓	↓	↓	↓
R4	↓	18	33	↓	↓	↓	↓

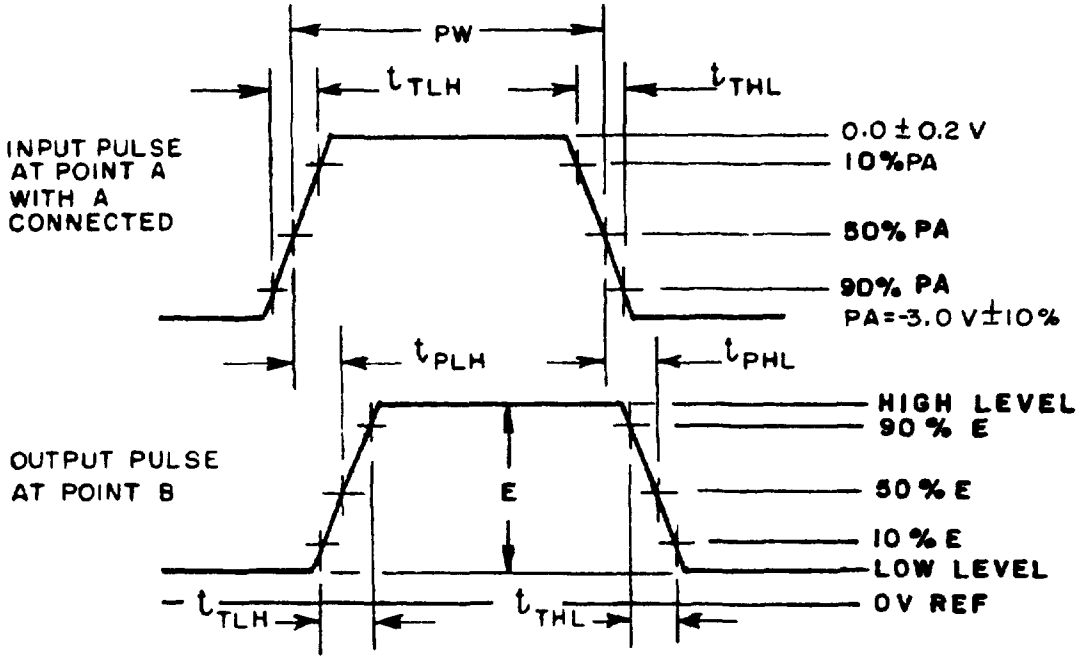
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NOTE: R1 = 51Ω ±5%, 1 W.

* FIGURE 13. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for table XII.

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NOTES:
1. Input pulse parameters:
   PW = 1 µs ±10%.
   PRR = 0.5 MHz ±10%.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. Before making measurements, the output levels shall be verified to be 0.4 V maximum for the low level and 2.4 V minimum for the high level.

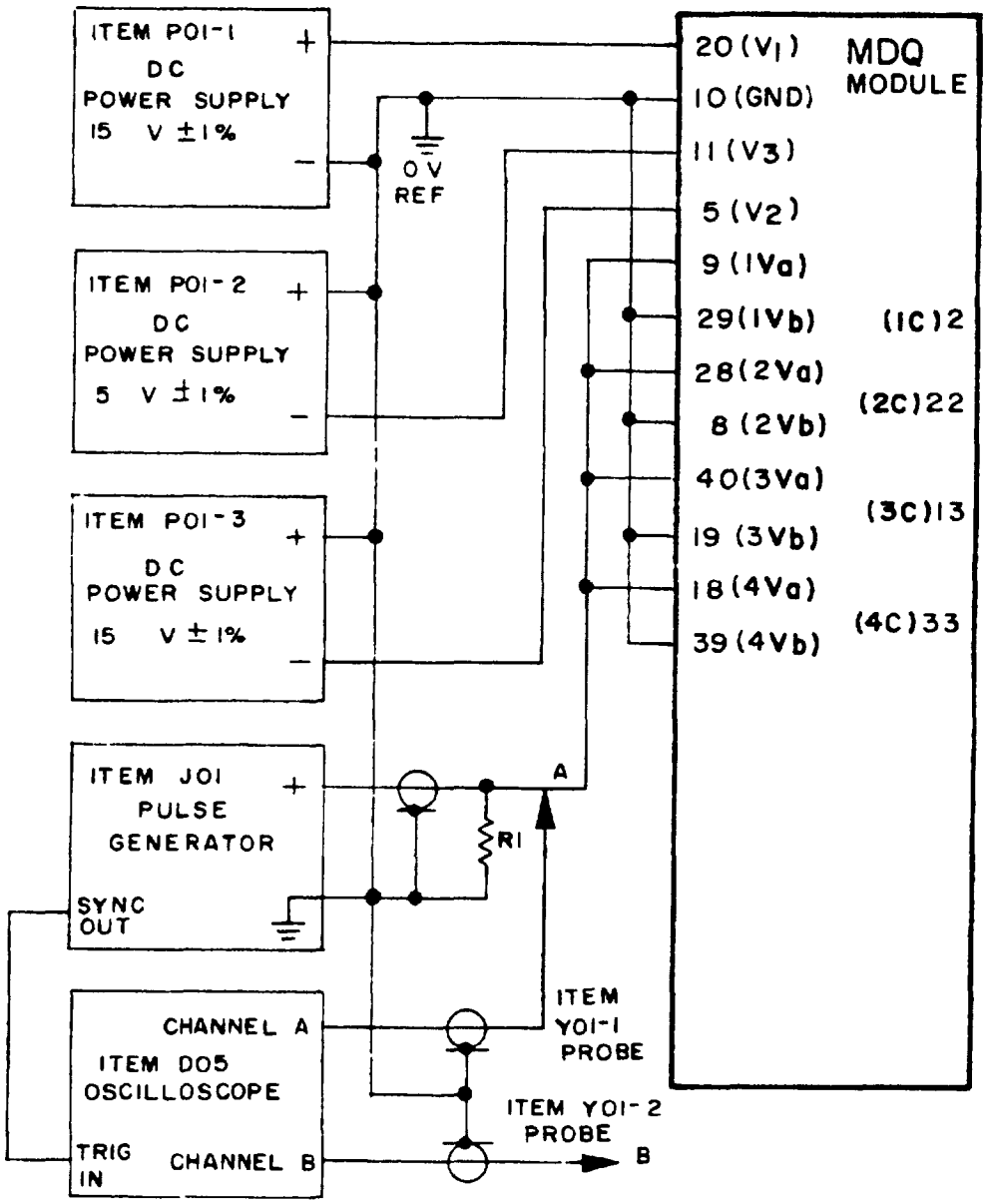
* FIGURE 14. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table XII.

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TABLE XIII. Power supply transient amplitude test.

Test condition	Parameter	Connect	Initial limits 25°±5°C ambient		Unit	Test figure
		B to pin	Min	Max		
S1	Power supply transient amplitude	20	0	400	mV	15,16
S2	Power supply transient amplitude	11	0	400	mV	15,16
S3	Power supply transient amplitude	5	0	400	mV	15,16

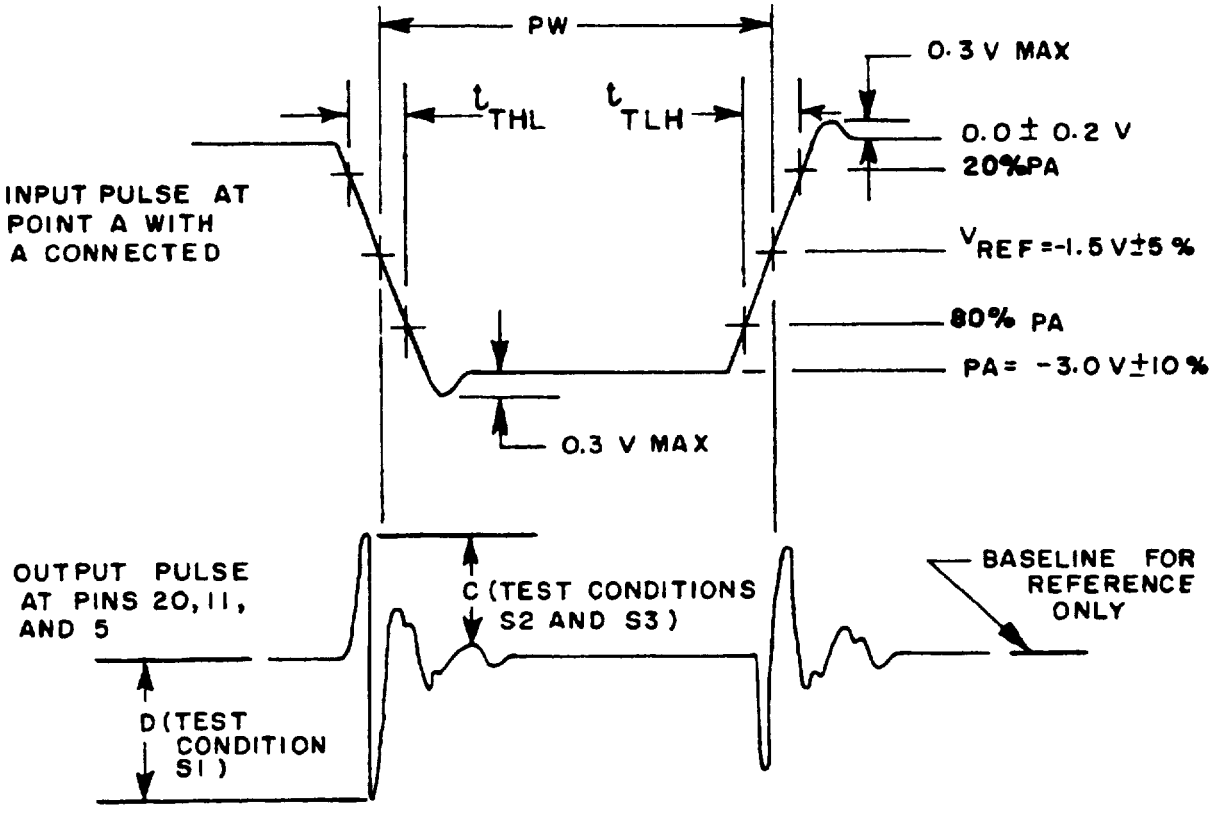
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NOTE: R1 = 51Ω ±5%, 1 W

FIGURE 15. Power supply transient amplitude test circuit for table XIII.

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NOTES:
1. Input pulse parameters.
   PW = 1 µs ±10%.
   PRR = 0.5 MHz ±10%
   $t_{TLH} = t_{THL}$ = 8 ±1 ns.
2. Amplitude D is measured from the baseline to the most negative excursion of the pulse.
3. Amplitude C is measured from the baseline to the most positive excursion of the pulse.

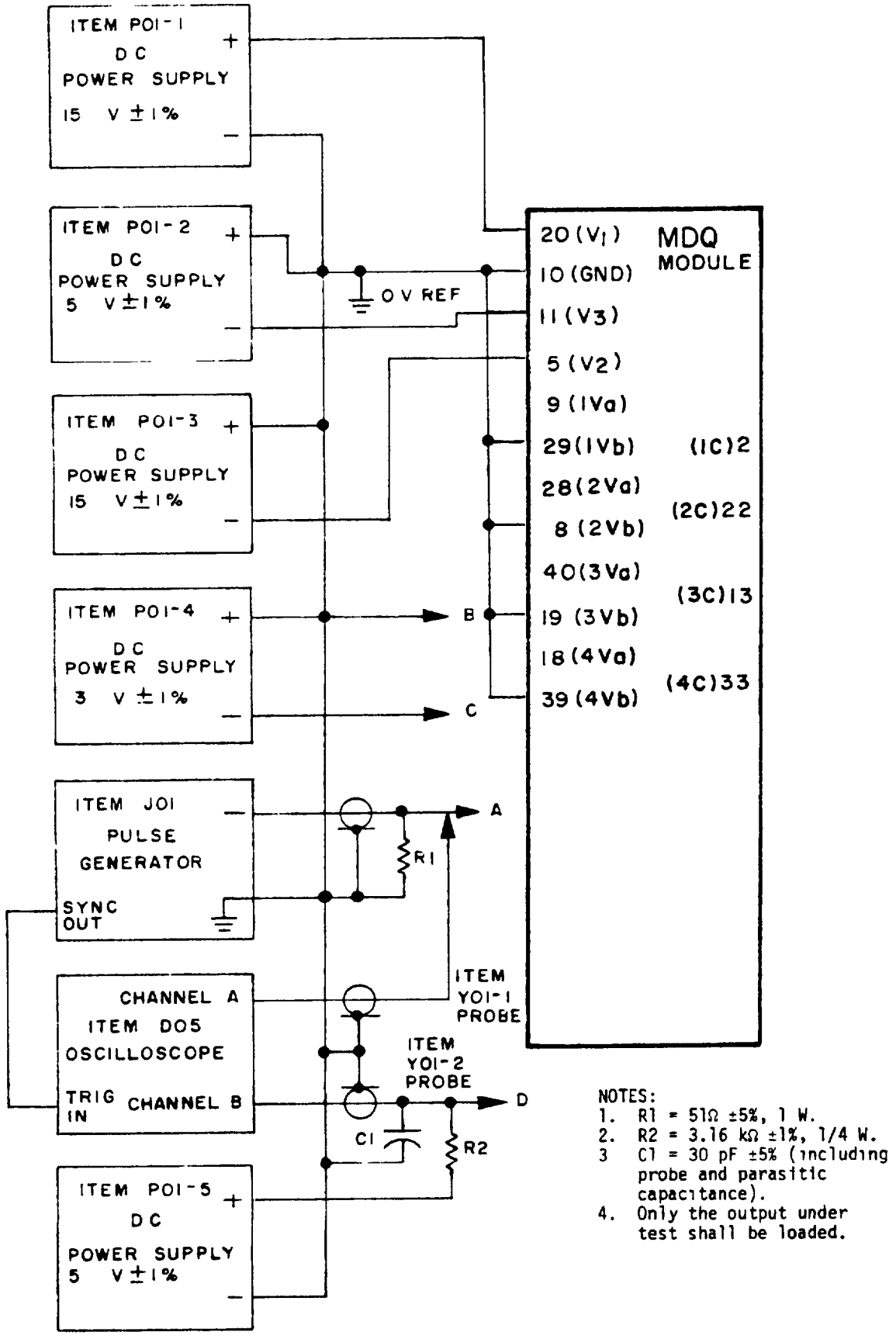
FIGURE 16. Power supply transient amplitude test circuit waveforms for table XIII.

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TABLE XIV. Output crosstalk (low-level and high-level) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
T1	Output crosstalk, low level	18,28, 40	Open	9	2	0	400	mV	17,18
T2		9,18, 40		28	22				
T3		9,18, 28		40	13				
T4		9,28, 40		18	33				
U1	Output crosstalk, high level	18,28, 40	9	Open	2				
U2		9,18, 40	28		22				
U3		9,18, 28	40		13				
U4		9,28, 40	18		33				

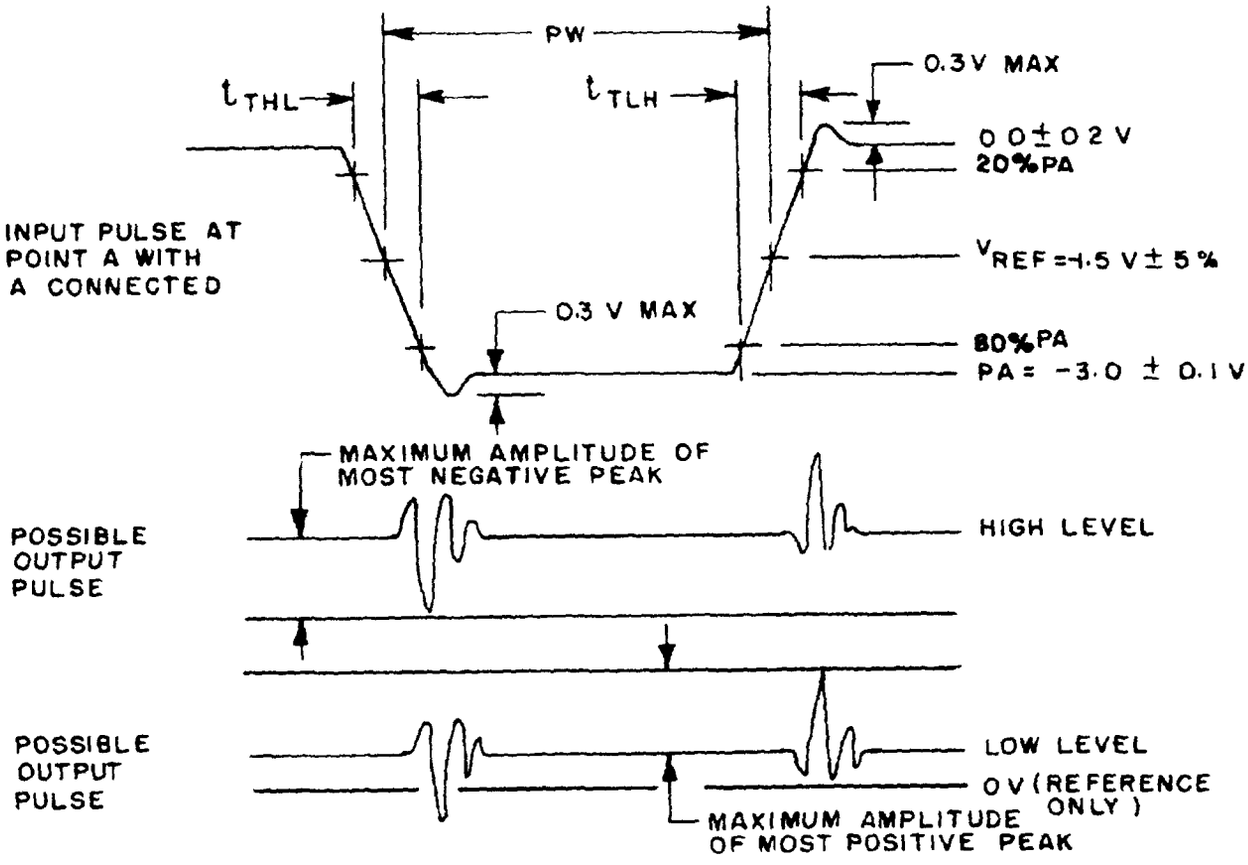
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NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. R2 = 3.16 kΩ ±1%, 1/4 W.
3 C1 = 30 pF ±5% (including probe and parasitic capacitance).
4. Only the output under test shall be loaded.

* FIGURE 17. Output crosstalk (low-level and high-level) test circuit for table XIV.

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NOTE: Input pulse parameters.
PW = 1 µs ±10%.
PRR = 0.5 MHz ±10%.
$t_{TLH} = t_{THL}$ = 8 ±2 ns.

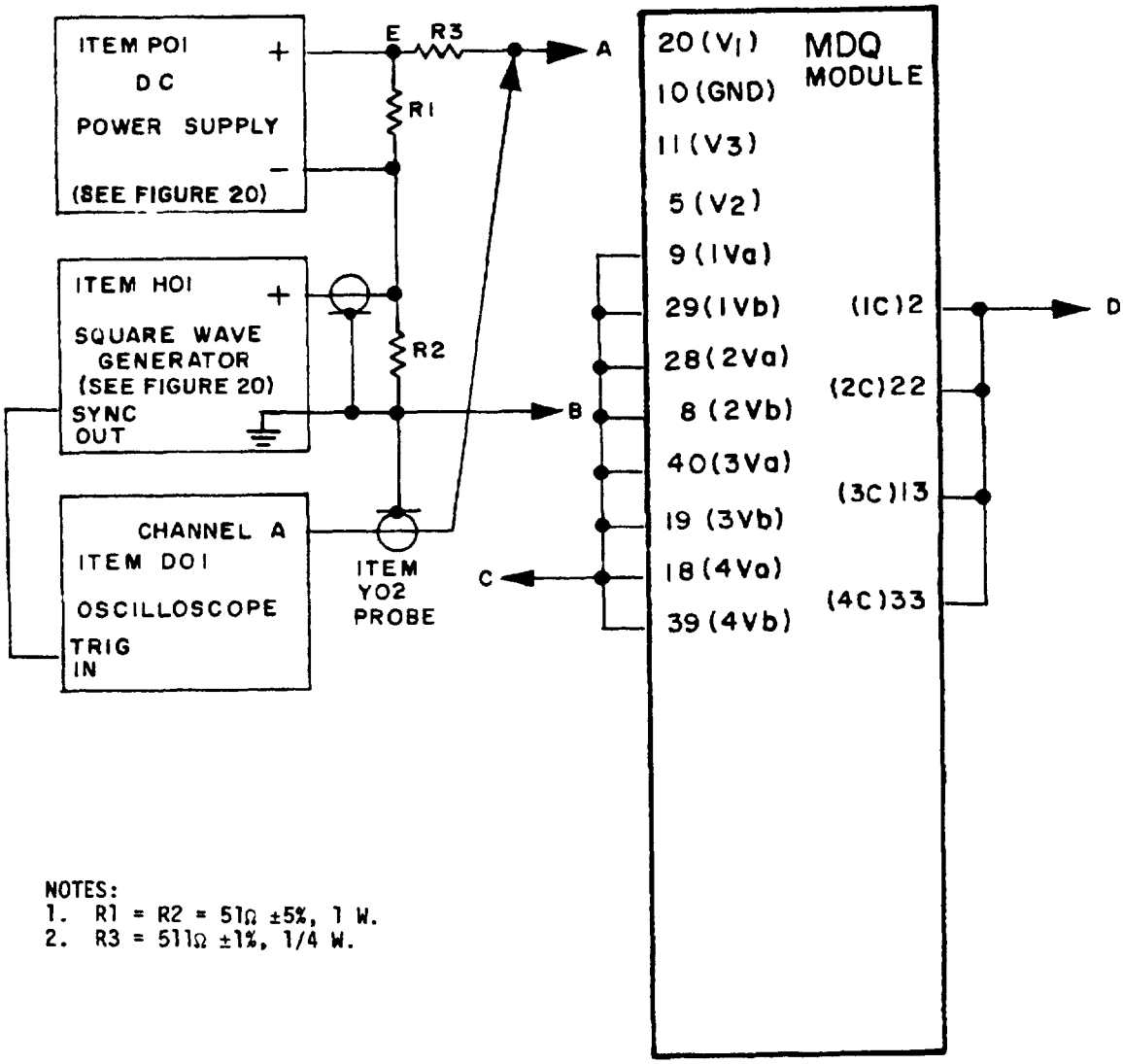
FIGURE 18. Output crosstalk (low-level and high-level) test circuit waveforms for table XIV.

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TABLE XV. Power supply transient rise time test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
V1	Power supply transient rise time	20	10	Open	Open	1.4	2.5	ms	19,20
V2	Power supply transient rise time	10,D	11	Open	A	5	12	µs	19,20
V3	Power supply transient rise time	10,11, 20, C,D	5	A	A	1.2	2.0	ms	19,20

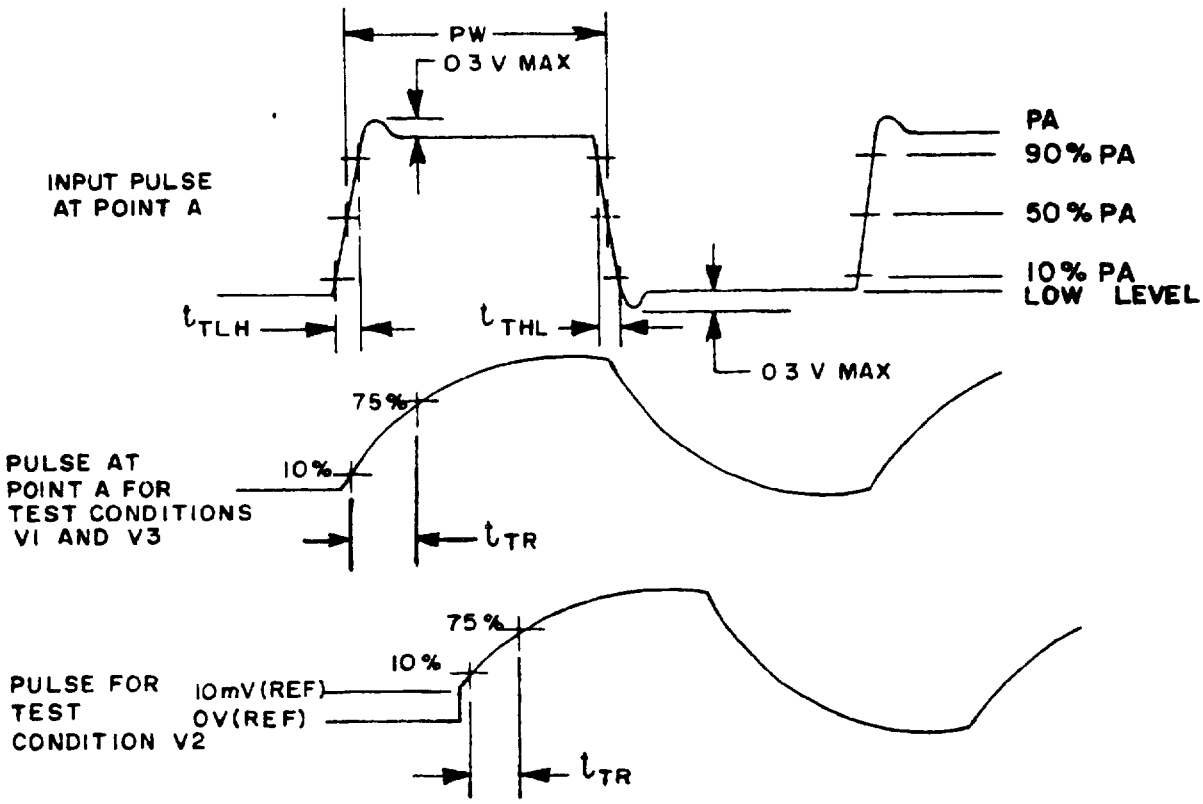
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1 W.
2. R3 = 511Ω ±1%, 1/4 W.

FIGURE 19 Power supply transient rise time test circuit for table XV.

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NOTES:
1. Test condition V1:

   a. PA = 0.5 V_{p-p} ±1%.
   b. PW = 10 ms ±10%.
   c. PRR = 50 Hz (reference).
   d. $t_{TLH} = t_{THL} < 100$ ns (reference).
   e. With point A open, adjust input as shown at point E.
   f. With point A open, adjust item P01 such that the low level of the input is at 0.00 ±0.01 V.
   g. Connect points A and B to pins specified in table XV, test condition V1; recheck low level and pulse parameters at point E.
   h. Measure t_{TR} at point A; the value measured shall comply with test condition V1.

* FIGURE 20. Power supply transient rise time test circuit waveforms for table XV.

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2 Test condition V2.

a PA = 0.25 V_{p-p} ±1%.
b PW = 100 µs ±10%
c. PRR = 5.0 kHz (reference).
d. $t_{TLH} = t_{THL}$ < 100 ns (reference).
e With point A open, adjust input as shown at point E
f With point A open, adjust item P01 such that the low level of the input is 0.00 ±0.01 V.
g. Connect points A, B, and D to pins specified in table XV , test condition V2, recheck low level and pulse parameters at point E.
h Measure t_{TR} at point A, the value measured shall comply with test condition V2

3. Test condition V3

a PA = 0 25 V_{p-p} ±1%.
b PW = 10 ms ±10%.
c. PRR = 50 Hz (reference).
d $t_{TLH} = t_{THL}$ < 100 ns (reference).
e. With point A open, adjust input as shown at point E.
f With point A open, adjust item P01 such that the low level of the input is at 0.00 ±0 01 V
g. Connect points A, B, C, and D to pins specified in table XV , test condition V3, recheck low level and pulse parameters at point E.
h. Measure t_{TR} at point A, the value measured shall comply with test condition V3.

* FIGURE 20. Power supply transient rise time test circuit waveforms for table XV — Continued.

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